

Striver's A2Z DSA Sheet

Complete Exhaustive Roadmap (Steps 1-18)

Total Problems: 455
Source: Codolio/TakeUForward

Step 1: Learn the Basics

Lec 1-6: Fundamentals, Patterns & Hashing

- ☐ User Input/Output & Data Types
- ☐ Pattern Printing (1-22 variations)
- ☐ Palindrome / GCD / Armstrong
- ☐ Recursion Basics (Name, 1-N, N-1)
- ☐ String Palindrome / Fibonacci
- ☐ Highest/Lowest Frequency Element
- ☐ If Else / Switch / Loops
- ☐ Count Digits / Reverse Number
- ☐ Print All Divisors / Prime Check
- ☐ Factorial / Reverse Array (Rec)
- ☐ Hashing: Counting Frequencies

Step 2: Sorting Techniques

Lec 1-2: Basic & Advanced Sorting

- ☐ Selection Sort
- ☐ Insertion Sort
- ☐ Quick Sort
- ☐ Bubble Sort
- ☐ Merge Sort
- ☐ Recursive Bubble/Insertion Sort

Step 3: Arrays (Easy, Medium, Hard)

Lec 1: Easy

- ☐ Largest/Second Largest
- ☐ Remove Duplicates
- ☐ Move Zeroes to End
- ☐ Find Missing Number
- ☐ Number that appears once
- ☐ Check if Sorted & Rotated
- ☐ Left Rotate by 1 / D places
- ☐ Union/Intersection of Sorted
- ☐ Max Consecutive Ones
- ☐ Longest Subarray with Sum K

Lec 2: Medium

- ☐ Two Sum / Sort 0s, 1s, 2s
- ☐ Kadane's Algorithm
- ☐ Rearrange Array by Sign
- ☐ Longest Consecutive Sequence
- ☐ Rotate Matrix (90 deg)
- ☐ Majority Element ($>n/2$)
- ☐ Stock Buy and Sell
- ☐ Next Permutation / Leaders
- ☐ Set Matrix Zeroes
- ☐ Spiral Matrix / Subarray Sum K

Lec 3: Hard

- ☐ Pascal's Triangle
- ☐ 3 Sum / 4 Sum Problems
- ☐ Count Subarrays with XOR K
- ☐ Merge Sorted Array (No Space)
- ☐ Count Inversions / Reverse Pairs
- ☐ Majority Element II ($>n/3$)
- ☐ Largest Subarray with 0 Sum
- ☐ Merge Overlapping Intervals
- ☐ Find Repeating and Missing
- ☐ Max Product Subarray

Step 4: Binary Search

Lec 1-3: BS on 1D, 2D and Search Space

- ☐ BS Implementation / Lower-Upper Bound
- ☐ Single Element in Sorted
- ☐ Koko Bananas / Bouquet making
- ☐ Painter's Partition / Gas Station
- ☐ Row with Max 1s / 2D Matrix Search
- ☐ First/Last Pos / Search Rotated
- ☐ Peak Element / Square & Nth Root
- ☐ Aggressive Cows / Book Allocation
- ☐ Median of Sorted Arrays
- ☐ Median in Row-Sorted Matrix

Step 5: Strings (Easy & Medium)

Lec 1-2

- ☐ Remove Outermost Parentheses
- ☐ Longest Common Prefix / Isomorphic
- ☐ Frequency Sort / Max Nesting Depth
- ☐ Atoi (String to Integer)
- ☐ Beauty Sum of Substrings
- ☐ Reverse Words / Largest Odd Number
- ☐ Rotate Strings / Valid Anagram
- ☐ Roman to Int / Int to Roman
- ☐ Longest Palindromic Substring

Step 6: Linked List

Singly & Doubly LL (Easy to Hard)

- ☐ Insert/Delete (SLL & DLL)
- ☐ Middle Node / Detect Loop
- ☐ Palindrome LL / Odd Even LL
- ☐ Intersection Point / Add 2 Numbers
- ☐ Reverse in K-Groups
- ☐ Reverse SLL & DLL
- ☐ Starting Point of Loop
- ☐ Remove Nth Node from Back
- ☐ Sort LL (0,1,2) / Merge 2 Sorted
- ☐ Flattening / Clone with Random

Step 7: Recursion & Step 8: Bit Manipulation

Lec 1-3

- ☐ Subsequences / Combination Sum I & II
- ☐ N-Queens / Sudoku / Rat in Maze
- ☐ Count Set Bits / XOR from 1-N
- ☐ Min Bit Flips / Divide 2 Integers
- ☐ Subset Sum I & II / Permutations
- ☐ Bit: Check set bit / Power of 2
- ☐ Two Odd Occurring / Power Set
- ☐ Sieve of Eratosthenes / Prime Factors

Step 9: Stack & Queues & Step 10: Sliding Window

Implementation & Monotonic Pattern

- ☐ Stack/Queue using Arrays & SLL
- ☐ Infix/Prefix/Postfix Conversions
- ☐ Trapping Rain Water / Histogram
- ☐ Max Consecutive Ones III
- ☐ Binary Subarrays Sum / Nice Subarrays
- ☐ Valid Parentheses / Min Stack
- ☐ Next Greater Element I & II
- ☐ LRU & LFU Cache
- ☐ Fruit into Baskets / Substrings with K
- ☐ Longest Repeating Substring (Replace)

Step 11: Heaps, Step 12: Greedy, Step 14: BST

Critical Patterns

- ☐ Heapify / Kth Largest / Kth Smallest
- ☐ Assign Cookies / Fractional Knapsack
- ☐ N Meetings / Candy / Jump Game I & II
- ☐ Kth Smallest/Largest in BST
- ☐ BST from Preorder / Two Sum BST
- ☐ Task Scheduler / Median in Stream
- ☐ Lemonade Change / Job Sequencing
- ☐ Search/Insert in BST / Floor-Ceil
- ☐ Validate BST / LCA in BST

Step 13: Trees, Step 15: Graphs, Step 16: DP

Advanced Data Structures

- ☐ Traversals / Balanced / Diameter
- ☐ BFS-DFS / Cycle Detection / Bipartite
- ☐ SCC (Kosaraju) / Articulation Points
- ☐ DP: Stocks / LIS / MCM / Partition
- ☐ Advanced Strings: Z-Function / KMP
- ☐ Vertical / View / LCA / Max Path Sum
- ☐ Dijkstra / Bellman-Ford / Prim-Kruskal
- ☐ DP: Grid / Knapsack / Coins / LCS
- ☐ Trie: Insert/Search / Longest Prefix

Complete Problem List for Striver's A2Z DSA Sheet | Track your journey to FAANG.